

TECHNOLOGY BRIEF · COMPANY 1 OF 2

Stateful *Swarms*

Iqidis & the irys-stateful-swarms engine — a persistent, multi-agent architecture for analyzing complex document sets, and how it could plug into DealDiligence.

COMPANY

Iqidis (iqidis.ai)

PRODUCT

Irys legal-AI platform

CODE

MIT · github.com/dl1683

EXECUTIVE SUMMARY

What this is, and why it matters to us

irys-stateful-swarms is an open-source (MIT) engine from **lqidis** — the team behind **Irys**, a legal-AI platform. It coordinates several inexpensive AI models around a shared, persistent "**blackboard**" of typed, source-cited findings that accumulates across iterations and sessions. The headline result: on a 1,251-task legal benchmark it beat both Harvey and frontier models on accuracy while costing **~40× less per task** — using cheap models that score 0% on their own.

17.75%

Strict all-pass on the Harvey Legal benchmark — vs Harvey's best 10.4% and frontier 13.3%

\$1.30

Cost per task vs Harvey's ~\$50.90 — about 40× cheaper

98×

"Intelligence per dollar" vs Harvey's initial result

WHY IT'S DIRECTLY RELEVANT

Its proven tasks *are* our domain — comparing a draft credit agreement to its commitment letter, flagging merger-remedy deviations against precedent, extracting liens and transaction entities. This is a more advanced, **stateful** version of exactly what DealDiligence does today (extract → reconcile → brief). And it ships an **MCP server** and already supports the Anthropic SDK, so it can plug into our Claude-native stack with minimal glue.

Bottom line up front: the architecture is adoptable and the code is permissively licensed. The highest-value ideas for us are (1) the persistent blackboard to upgrade our single-pass reconciliation, (2) a cheap-model cascade to cut our biggest cost line — extraction, and (3) persistent "matter memory" that makes repeat questions on a deal nearly free. Recommended first step: a low-cost **MCP pilot on 2–3 real deals**. Caveats and risks are covered on the final pages.

01 · THE COMPANY & THE CODE

Who's behind it

Iqidis — the company

A small team building **Irys** (irys.ai), a "unified legal AI platform" positioned against Harvey. The commercial product adds persistent document indexes, entity & knowledge graphs, citation verification against **50M+ court opinions**, drafting with tracked changes, and a matter-management workspace. Contact is public (devansh@iqidis.ai); they run a monthly contributor bounty and recruit from the open-source work.

irys-stateful-swarms — the open repo

The **coordination engine** open-sourced under **MIT** — a Python 3.12 package. It deliberately ships the *stateless* version (every task starts from zero) so the benchmark is fair; the persistent layer lives in the commercial product and a companion repo (MapU). Multi-provider: Google Gemini (default), **Anthropic**, and OpenAI. Includes a CLI and an **MCP server**.

NAME COLLISION – IMPORTANT FOR DUE DILIGENCE

This "**Irys**" (legal AI, irys.ai, by Iqidis) is **unrelated** to "**Irys**" the crypto/blockchain "programmable datachain" (irys.xyz). Same name, different companies. If you search the web you'll mostly hit the blockchain one — ignore it; everything in this brief is the legal-AI Iqidis project.

The broader open-source stack

The swarm engine is one of five pieces Iqidis has released. The others are research-grade but point at problems we also have:

COMPONENT	WHAT IT DOES	RELEVANCE TO US
irys-stateful-swarms	Multi-agent coordination + blackboard state	Core — our extraction/reconciliation analogue
MapU	Persistent, conflict-aware knowledge memory (Postgres + pgvector)	Our planned "pgvector retriever," done
Fractal Embeddings	Hierarchical embeddings (domain → category → detail)	Better clause-in-context retrieval
Latent Space Reasoning	Improves a model's reasoning at inference, no training	Cheaper accuracy on weak stages
CTI Universal Law	Predicts which model fits which task	Principled model routing

02 · THE IDEA

The problem it solves: forgetfulness

Today's AI systems forget everything between sessions. Every question re-reads the same documents, re-discovers the same entities, and re-derives the same analysis from scratch. Context windows get compacted, destroying detail; session boundaries erase progress; ordinary retrieval pulls back text snippets but not the *understanding* built from them.

The fix: a persistent "blackboard"

Instead of treating each analysis as disposable, the system writes every finding to a shared, structured **blackboard** — a typed, fully source-attributed knowledge base. Every observation, calculation, analysis, and *gap* is preserved with a pointer back to the exact document and section it came from. Nothing is summarized away; nothing is forgotten. The blackboard — not the final memo — is the real artifact: an inspectable, auditable reasoning trace that accumulates over time and makes each subsequent question cheaper and more accurate than the last.

THE THESIS THAT MAKES IT REMARKABLE

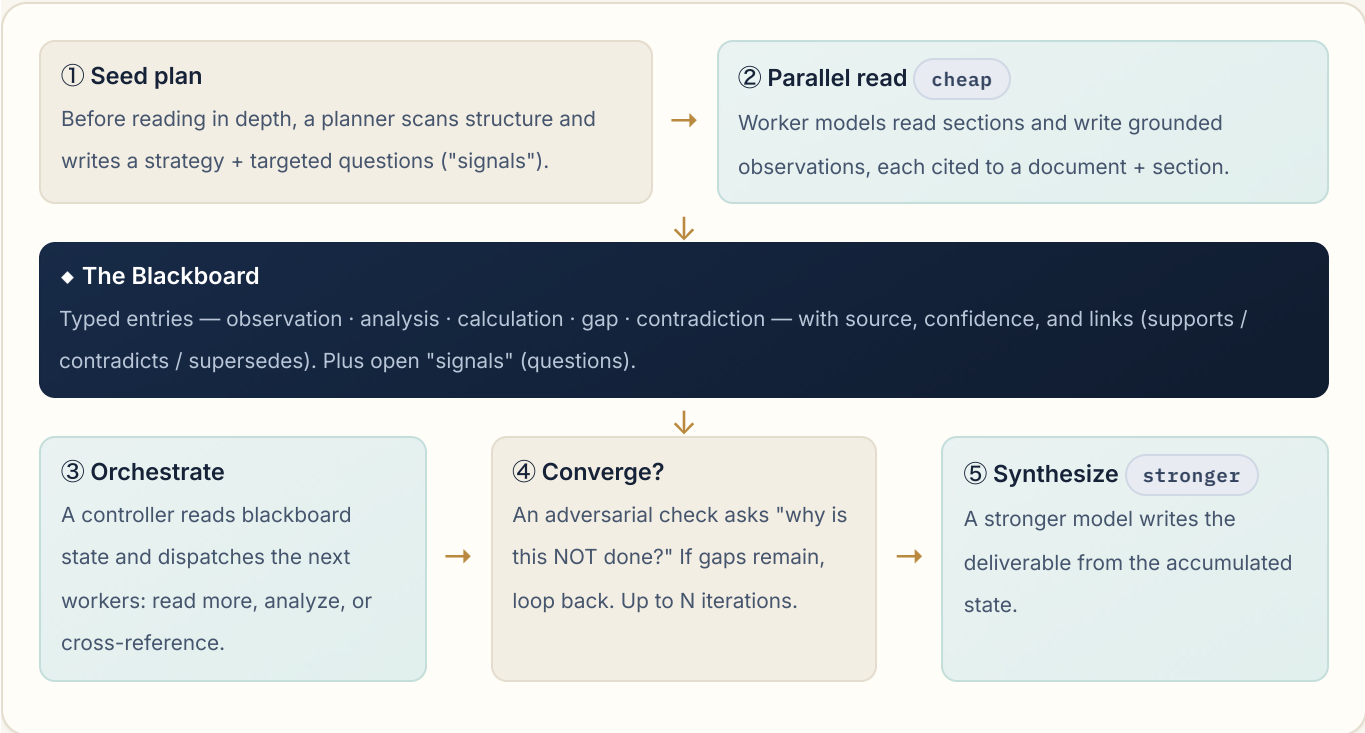
The same cheap Gemini models that score **0%** in other agent frameworks hit **17.75%** here. The conclusion the authors draw — and demonstrate — is that **the capability comes from the architecture, not the model**. You don't need a frontier model to do professional document analysis; you need a system that knows how to build and maintain analytical state.

For us, the economic punchline matters most: extraction is ~70% of the cost and is paid **once** per document set in a stateful deployment. Follow-up questions skip re-reading entirely. Iqidis claims their persistent product cuts multi-turn cost by up to **1,000×** versus re-computing from scratch.

03 · UNDER THE HOOD

How the swarm actually works

It behaves less like a pipeline and more like a **control system**: it measures the state of its own understanding, compares it against what the task demands, and dispatches small workers to close the gaps — looping until measurable conditions are met.



The three-tier model cascade

Work is routed to the cheapest model that can do it — the cost lever that makes the economics work.

TIER	JOB	DEFAULT MODEL
Read	Source reading, extraction, structural profiling	Gemini 3.1 Flash Lite (\$0.25/M)
Reason	Analysis, cross-reference, gap detection, synthesis	Gemini 3.5 Flash (\$1.50/M)
Construct	Supervisor review, seed planning, hard judgment	A "Pro/Opus" tier when available

The models are swappable by env var — our own Claude tiers (Haiku → Sonnet → Opus) map cleanly onto Read → Reason → Construct.

04 · THE QUALITY MACHINERY

How it polices its own work

This is the part most aligned with our biggest differentiator — the verify layer. The swarm treats **"matter custody"** as a first-class duty: preserving the integrity of every fact from source to final artifact.

Signals & gaps

Open questions ("signals") and explicit **gap** entries keep a running list of what's still unanswered. A worked banking example grew from **7 entries to 2,400** grounded findings over 12 iterations — a 343× expansion — with 135 gaps logged, not hidden.

Debt sensors

Optional detectors flag specific failure modes: **relation debt** (a needed cross-document comparison wasn't done), **authority debt** (a claim lacks a citation), **severity debt** (a risk wasn't rated), **source-object debt** (an entity/population was missed).

The custody-break taxonomy — a diagnostic vocabulary for failures

They classify *how* information gets lost between source and output across 12 named types — e.g. absent-state (a fact never entered), unpromoted-fact (extracted but never analyzed), lost-commitment ("must calculate X" noted but never done), false-completion (declared done with gaps open). Their own analysis: in nearly every near-miss, **the right fact was already on the blackboard** — the missing step was the final cross-reference or citation. That's a state-processing fix, not a model-intelligence ceiling.

CONCRETE PROOF POINT — AN M&A TASK

On an antitrust merger-remedies comparison it passed **56 of 61** criteria — correctly extracting that a divestiture proposed transferring only **14 FTEs vs ~85** in precedent, a **3-year supply agreement vs 5**, an off-market **4.5% royalty vs a 2–3% norm**, and a monitoring-trustee conflict of interest — each cited to specific cases. On a credit-agreement-vs-commitment-letter comparison it scored a perfect **40/40**.

05 · THE FIT

How it maps onto DealDiligence

Almost every stage of our pipeline has a richer counterpart in the swarm model. This is less "a new tool" and more "a more advanced version of our own architecture."

DEALDILIGENCE TODAY	SWARM COUNTERPART	UPGRADE
Per-document fact sheet (one Opus read)	Blackboard observations — typed, cited, confidence-scored	Iterative & gap-aware vs one-shot
Reconciliation (single Sonnet pass, ≥2 docs)	Iterative cross-document convergence driven by signals	Catches deviations one pass misses
Anomaly detection (≥3 docs)	Relation-debt & deviation analysis	The credit-agreement example <i>is</i> this
Verify layer (citation check + Sonnet)	Matter custody + custody-break taxonomy + authority debt	Deeper provenance & fewer misses
Deal brief (Sonnet synthesis)	Synthesis from curated state + "must-surface" policy	Values/contradictions never dropped
Kanban risk report (Opus)	Swarm "draft / analyze" task (90%+ on draft)	Same job, cheaper & auditable
Planned pgvector retriever	MapU persistent memory (Postgres + pgvector)	Already built, conflict-aware
Opus-tier extraction cost (~70% of spend)	Flash/Haiku-Lite read cascade	Large cost reduction

THE ONE GAP THEY HAVEN'T SOLVED — THAT WE ALSO HAVE

Their top open research question is **cross-document entity resolution without LLM calls** ("Acme Corp" vs "Acme Corporation") — which is precisely what our reconciliation's MasterEntity merge tackles. We are not behind here; if anything it's a place we could contribute or compare notes.

06 • WHAT WE COULD ACTUALLY DO

Potential implementations

Ranked roughly by effort-to-value. Each carries an effort and an impact rating.

1

Pilot it via MCP

Effort: Low

Signal: High

They ship an **MCP server** and DealDiligence is Claude-native — MCP is a first-class integration path. Wire the swarm in as a tool, point it at 2–3 real deals' document sets, and A/B its output against our current extraction + reconciliation. Fastest, cheapest way to validate the claims on *our* data before committing to anything. Configure its model env vars to use our Anthropic keys.

2

Adopt the blackboard for reconciliation

Effort: Med

Impact: High

Replace our single-pass reconciliation with an iterative, gap-driven loop over a typed blackboard. Our cross-document deviation detection (the heart of diligence) gets materially stronger — the merger-remedies and credit-agreement examples show the depth a converging loop reaches versus one shot. Keep our existing fact sheets as the seed observations.

3

Cut extraction cost with a cheap-model cascade

Effort: Med

Impact: High

Extraction is our biggest cost line (~70%). The swarm's core claim is that architecture, not model size, drives quality — cheap readers + coordination beat lone frontier models. Moving first-pass reading from Opus onto a Haiku/Flash-Lite cascade, with the swarm recovering quality, directly attacks our unit economics. Validate against our verify layer to confirm no accuracy loss.

4

Persistent matter memory (MapU)

Effort: Med

Impact: High (strategic)

Adopt MapU-style persistent, conflict-aware state so the 10th question on a deal costs a fraction of the first, and a new document reconciles *incrementally* against existing understanding rather than triggering a full rebuild. This is the "pgvector retriever" already on our roadmap — built, and with provenance + supersession baked in.

5

Harden our verify layer with debt sensors

Effort: Med

Impact: Med

Port the custody-break taxonomy and authority/relation debt checks into our verification step to cut "needs review" misses — the cases where the fact was found but the cross-reference or citation wasn't made. Strengthens the SOC 2 / regulated-buyer story that is core to our positioning.

6

Hierarchical retrieval (Fractal Embeddings)

Effort: High

Impact: Med

Longer-horizon: embeddings that natively understand "this SOFR-floor clause lives in a credit-agreement section, in a banking deal" improve cross-reference detection — exactly where near-misses happen. Pairs with our eventual move off the stuff-everything retriever.

07 · THE HONEST ASSESSMENT

Engagement options & risks

Adopt the code

MIT license — use the swarm engine and MapU directly, no contract. Lowest friction, full control, but we own the integration and maintenance.

Partner with Iqidis

Their commercial Irys adds citation-vs-50M-opinions and matter management. But Irys is a *legal-AI product* in an adjacent lane — partner vs. competitor needs judgment.

Talent channel

Active team, public contact, bounty program. A pragmatic way to engage the authors or recruit the expertise if we build our own.

Risks & caveats — read before betting on it

- **Early-stage by their own admission.** The design doc is labeled "PHASE 0 READY; FULL DESIGN UNPROVEN." The debt sensors default **off** and their precision/recall are explicitly *unvalidated*. The advanced custody machinery is largely roadmap, not shipped.
- **Confidence scores aren't calibrated.** They state plainly that confidence values are model heuristics, not probabilities — don't build reliability claims on them.
- **A known architectural hole.** "Late-lifecycle mutation": sensors can add findings *after* the convergence check, so the approved state can be stale at synthesis. Documented, not yet fixed.
- **Entity resolution isn't solved** — it's their #1 open question. Their near-misses repeatedly stem from unreconciled duplicate entities, the same problem our reconciliation addresses.
- **Benchmark fine print.** Run on the *public* Harvey set (not Harvey's private holdout) and judged with a Gemini model rather than the recommended Sonnet (they report ~90% agreement). Impressive and transparent, but not an apples-to-apples Harvey comparison.
- **Gemini-centric defaults.** Anthropic is supported, but the tuning and benchmarks are Gemini-Flash-based; expect calibration work to hit parity on Claude.

RECOMMENDATION

Treat it as a **high-potential reference architecture, not a turnkey dependency**. Run the low-cost MCP pilot on real deals to validate on our data. If it holds up, the blackboard-driven reconciliation and persistent matter memory are the two ideas most likely to move our accuracy and cost at once. Keep Iqidis on the radar as both a partnership option and a competitive datapoint — and don't confuse them with the blockchain "Irys."

Sources: the project README, `docs/SWARM_INTELLIGENCE.md`, source (`src/swarm/`, `.mcp.json`, `pyproject.toml`), and the included M&A example outputs — github.com/dl1683/irys-stateful-swarms. Company 2 of 2 to follow.